

## VESSEL ANALYSIS REPORT

| vaf(%) | branch points count | tip point count | vein count | total vein length | average vein length |
|--------|---------------------|-----------------|------------|-------------------|---------------------|
| 1.52   | 69.0                | 172.0           | 189.0      | 4358.757          | 23.062              |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 1  | [54,11]    | [68,11]    | 13.0   | BRANCH  | BRANCH  |
| 2  | [54,11]    | [18,28]    | 43.284 | BRANCH  | BRANCH  |
| 3  | [54,11]    | [53,37]    | 27.485 | BRANCH  | BRANCH  |
| 4  | [79,1]     | [68,11]    | 13.728 | TIP     | BRANCH  |
| 5  | [91,16]    | [68,11]    | 26.142 | TIP     | BRANCH  |
| 6  | [91,0]     | [131,11]   | 43.556 | TIP     | BRANCH  |
| 7  | [131,11]   | [138,20]   | 11.485 | BRANCH  | BRANCH  |
| 8  | [137,2]    | [131,11]   | 10.657 | TIP     | BRANCH  |
| 9  | [374,1]    | [321,12]   | 56.556 | TIP     | BRANCH  |
| 10 | [306,19]   | [321,12]   | 18.899 | TIP     | BRANCH  |
| 11 | [318,63]   | [321,12]   | 52.899 | TIP     | BRANCH  |
| 12 | [485,13]   | [482,18]   | 5.243  | TIP     | BRANCH  |
| 13 | [449,22]   | [482,18]   | 36.728 | TIP     | BRANCH  |
| 14 | [484,85]   | [482,18]   | 92.225 | TIP     | BRANCH  |
| 15 | [138,20]   | [147,20]   | 8.0    | BRANCH  | BRANCH  |
| 16 | [138,20]   | [155,35]   | 37.314 | BRANCH  | BRANCH  |
| 17 | [148,18]   | [147,20]   | 1.414  | TIP     | BRANCH  |
| 18 | [147,20]   | [155,35]   | 16.899 | BRANCH  | BRANCH  |
| 19 | [0,1]      | [15,27]    | 36.485 | TIP     | BRANCH  |
| 20 | [15,27]    | [6,65]     | 43.657 | BRANCH  | BRANCH  |
| 21 | [15,27]    | [18,28]    | 2.0    | BRANCH  | BRANCH  |
| 22 | [18,28]    | [22,31]    | 3.828  | BRANCH  | BRANCH  |
| 23 | [33,32]    | [22,31]    | 10.414 | TIP     | BRANCH  |
| 24 | [22,31]    | [28,56]    | 26.485 | BRANCH  | BRANCH  |
| 25 | [155,35]   | [173,41]   | 19.485 | BRANCH  | BRANCH  |
| 26 | [53,37]    | [68,38]    | 14.414 | BRANCH  | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 27 | [53,37]    | [42,72]    | 47.142 | BRANCH  | BRANCH  |
| 28 | [68,38]    | [68,42]    | 3.0    | BRANCH  | BRANCH  |
| 29 | [70,35]    | [68,38]    | 2.414  | TIP     | BRANCH  |
| 30 | [193,62]   | [173,41]   | 30.042 | TIP     | BRANCH  |
| 31 | [186,2]    | [173,41]   | 47.213 | TIP     | BRANCH  |
| 32 | [68,42]    | [78,42]    | 9.0    | BRANCH  | BRANCH  |
| 33 | [68,42]    | [74,65]    | 24.485 | BRANCH  | BRANCH  |
| 34 | [80,45]    | [78,42]    | 2.828  | TIP     | BRANCH  |
| 35 | [78,42]    | [105,47]   | 32.728 | BRANCH  | BRANCH  |
| 36 | [120,48]   | [105,47]   | 16.071 | TIP     | BRANCH  |
| 37 | [93,54]    | [105,47]   | 13.485 | TIP     | BRANCH  |
| 38 | [105,47]   | [123,67]   | 29.556 | BRANCH  | BRANCH  |
| 39 | [28,56]    | [42,72]    | 21.385 | BRANCH  | BRANCH  |
| 40 | [24,58]    | [28,56]    | 3.828  | TIP     | BRANCH  |
| 41 | [6,65]     | [15,75]    | 13.899 | BRANCH  | BRANCH  |
| 42 | [1,72]     | [6,65]     | 7.657  | TIP     | BRANCH  |
| 43 | [74,65]    | [79,65]    | 4.0    | BRANCH  | BRANCH  |
| 44 | [74,65]    | [55,86]    | 27.456 | BRANCH  | BRANCH  |
| 45 | [86,58]    | [79,65]    | 9.485  | TIP     | BRANCH  |
| 46 | [98,83]    | [79,65]    | 26.627 | TIP     | BRANCH  |
| 47 | [130,110]  | [123,67]   | 62.456 | TIP     | BRANCH  |
| 48 | [109,81]   | [123,67]   | 20.556 | TIP     | BRANCH  |
| 49 | [42,72]    | [44,86]    | 13.828 | BRANCH  | BRANCH  |
| 50 | [299,65]   | [280,72]   | 20.899 | TIP     | BRANCH  |
| 51 | [275,100]  | [280,72]   | 29.899 | TIP     | BRANCH  |
| 52 | [278,68]   | [280,72]   | 3.414  | TIP     | BRANCH  |
| 53 | [15,75]    | [37,93]    | 29.627 | BRANCH  | BRANCH  |
| 54 | [11,94]    | [15,75]    | 19.243 | TIP     | BRANCH  |
| 55 | [155,86]   | [202,78]   | 53.456 | TIP     | BRANCH  |
| 56 | [170,96]   | [202,78]   | 44.314 | TIP     | BRANCH  |
| 57 | [195,46]   | [202,78]   | 35.142 | TIP     | BRANCH  |
| 58 | [44,86]    | [55,86]    | 10.0   | BRANCH  | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 59 | [44,86]    | [37,93]    | 9.485   | BRANCH  | BRANCH  |
| 60 | [62,116]   | [55,86]    | 32.314  | TIP     | BRANCH  |
| 61 | [37,93]    | [35,106]   | 12.828  | BRANCH  | BRANCH  |
| 62 | [77,89]    | [78,104]   | 15.243  | TIP     | BRANCH  |
| 63 | [101,110]  | [78,104]   | 24.485  | TIP     | BRANCH  |
| 64 | [78,104]   | [131,129]  | 76.698  | BRANCH  | BRANCH  |
| 65 | [35,106]   | [43,106]   | 7.0     | BRANCH  | BRANCH  |
| 66 | [0,106]    | [35,106]   | 39.142  | TIP     | BRANCH  |
| 67 | [49,101]   | [43,106]   | 7.071   | TIP     | BRANCH  |
| 68 | [45,129]   | [43,106]   | 24.485  | TIP     | BRANCH  |
| 69 | [169,122]  | [131,129]  | 40.728  | TIP     | BRANCH  |
| 70 | [131,129]  | [146,168]  | 45.042  | BRANCH  | BRANCH  |
| 71 | [9,137]    | [16,147]   | 12.485  | TIP     | BRANCH  |
| 72 | [16,147]   | [34,159]   | 23.142  | BRANCH  | BRANCH  |
| 73 | [1,158]    | [16,147]   | 19.899  | TIP     | BRANCH  |
| 74 | [46,147]   | [46,156]   | 8.828   | TIP     | BRANCH  |
| 75 | [46,156]   | [34,159]   | 11.828  | BRANCH  | BRANCH  |
| 76 | [46,156]   | [66,158]   | 22.728  | BRANCH  | BRANCH  |
| 77 | [62,139]   | [66,158]   | 20.485  | TIP     | BRANCH  |
| 78 | [66,158]   | [70,177]   | 23.657  | BRANCH  | BRANCH  |
| 79 | [34,159]   | [26,189]   | 32.899  | BRANCH  | BRANCH  |
| 80 | [167,165]  | [158,168]  | 9.243   | BRANCH  | BRANCH  |
| 81 | [171,160]  | [167,165]  | 5.243   | TIP     | BRANCH  |
| 82 | [146,168]  | [158,168]  | 11.0    | BRANCH  | BRANCH  |
| 83 | [146,168]  | [107,175]  | 41.314  | BRANCH  | BRANCH  |
| 84 | [260,198]  | [158,168]  | 114.841 | TIP     | BRANCH  |
| 85 | [107,175]  | [70,177]   | 36.828  | BRANCH  | BRANCH  |
| 86 | [116,193]  | [107,175]  | 20.728  | TIP     | BRANCH  |
| 87 | [70,177]   | [57,190]   | 19.142  | BRANCH  | BRANCH  |
| 88 | [23,189]   | [26,189]   | 2.0     | BRANCH  | BRANCH  |
| 89 | [19,172]   | [23,189]   | 31.971  | TIP     | BRANCH  |
| 90 | [23,189]   | [7,219]    | 35.213  | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 91  | [26,189]   | [28,214]   | 25.657 | BRANCH  | BRANCH  |
| 92  | [39,190]   | [57,190]   | 18.657 | TIP     | BRANCH  |
| 93  | [73,204]   | [57,190]   | 20.971 | TIP     | BRANCH  |
| 94  | [35,214]   | [28,214]   | 6.0    | TIP     | BRANCH  |
| 95  | [28,214]   | [24,221]   | 7.243  | BRANCH  | BRANCH  |
| 96  | [4,220]    | [7,219]    | 2.414  | TIP     | BRANCH  |
| 97  | [7,219]    | [15,223]   | 9.243  | BRANCH  | BRANCH  |
| 98  | [24,221]   | [15,223]   | 8.828  | BRANCH  | BRANCH  |
| 99  | [24,221]   | [25,235]   | 13.828 | BRANCH  | BRANCH  |
| 100 | [15,223]   | [25,235]   | 16.899 | BRANCH  | BRANCH  |
| 101 | [87,229]   | [73,229]   | 13.0   | TIP     | BRANCH  |
| 102 | [55,217]   | [73,229]   | 21.556 | TIP     | BRANCH  |
| 103 | [73,229]   | [69,248]   | 19.243 | BRANCH  | BRANCH  |
| 104 | [306,235]  | [294,233]  | 11.828 | TIP     | BRANCH  |
| 105 | [291,238]  | [294,233]  | 5.828  | TIP     | BRANCH  |
| 106 | [267,204]  | [294,233]  | 39.941 | TIP     | BRANCH  |
| 107 | [25,235]   | [7,300]    | 74.527 | BRANCH  | BRANCH  |
| 108 | [211,242]  | [208,243]  | 2.414  | TIP     | BRANCH  |
| 109 | [206,242]  | [208,243]  | 1.414  | TIP     | BRANCH  |
| 110 | [208,245]  | [208,243]  | 1.0    | TIP     | BRANCH  |
| 111 | [69,248]   | [54,253]   | 16.071 | BRANCH  | BRANCH  |
| 112 | [75,278]   | [69,248]   | 34.971 | TIP     | BRANCH  |
| 113 | [46,279]   | [54,253]   | 28.314 | TIP     | BRANCH  |
| 114 | [48,251]   | [54,253]   | 5.414  | TIP     | BRANCH  |
| 115 | [459,252]  | [461,253]  | 1.414  | TIP     | BRANCH  |
| 116 | [469,274]  | [461,253]  | 23.314 | TIP     | BRANCH  |
| 117 | [511,220]  | [461,253]  | 64.598 | TIP     | BRANCH  |
| 118 | [137,251]  | [138,256]  | 4.414  | TIP     | BRANCH  |
| 119 | [154,262]  | [138,256]  | 17.485 | TIP     | BRANCH  |
| 120 | [130,275]  | [138,256]  | 21.314 | TIP     | BRANCH  |
| 121 | [2,299]    | [7,300]    | 4.414  | TIP     | BRANCH  |
| 122 | [0,314]    | [7,300]    | 15.899 | TIP     | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 123 | [130,288]  | [131,311]  | 23.243 | TIP     | BRANCH  |
| 124 | [140,315]  | [131,311]  | 9.657  | TIP     | BRANCH  |
| 125 | [131,311]  | [114,331]  | 28.971 | BRANCH  | BRANCH  |
| 126 | [170,328]  | [169,330]  | 1.414  | TIP     | BRANCH  |
| 127 | [178,339]  | [169,330]  | 12.899 | TIP     | BRANCH  |
| 128 | [166,331]  | [169,330]  | 2.0    | TIP     | BRANCH  |
| 129 | [114,331]  | [81,366]   | 50.598 | BRANCH  | BRANCH  |
| 130 | [115,244]  | [114,331]  | 95.113 | TIP     | BRANCH  |
| 131 | [71,353]   | [68,361]   | 8.243  | TIP     | BRANCH  |
| 132 | [1,340]    | [68,361]   | 75.527 | TIP     | BRANCH  |
| 133 | [68,361]   | [69,366]   | 4.414  | BRANCH  | BRANCH  |
| 134 | [69,366]   | [81,366]   | 11.0   | BRANCH  | BRANCH  |
| 135 | [69,366]   | [43,424]   | 67.355 | BRANCH  | BRANCH  |
| 136 | [128,347]  | [81,366]   | 72.012 | TIP     | BRANCH  |
| 137 | [23,422]   | [43,424]   | 19.828 | BRANCH  | BRANCH  |
| 138 | [20,437]   | [23,422]   | 15.243 | TIP     | BRANCH  |
| 139 | [1,428]    | [23,422]   | 25.899 | TIP     | BRANCH  |
| 140 | [43,424]   | [66,431]   | 24.899 | BRANCH  | BRANCH  |
| 141 | [308,429]  | [323,429]  | 15.657 | TIP     | BRANCH  |
| 142 | [325,438]  | [323,429]  | 8.828  | TIP     | BRANCH  |
| 143 | [355,420]  | [323,429]  | 34.314 | TIP     | BRANCH  |
| 144 | [68,428]   | [66,431]   | 2.828  | TIP     | BRANCH  |
| 145 | [151,448]  | [66,431]   | 98.012 | TIP     | BRANCH  |
| 146 | [171,505]  | [169,508]  | 2.414  | TIP     | BRANCH  |
| 147 | [157,510]  | [169,508]  | 11.414 | TIP     | BRANCH  |
| 148 | [173,509]  | [169,508]  | 3.0    | TIP     | BRANCH  |
| 149 | [209,0]    | [202,15]   | 17.899 | TIP     | TIP     |
| 150 | [351,14]   | [359,38]   | 34.485 | TIP     | TIP     |
| 151 | [395,25]   | [410,59]   | 43.87  | TIP     | TIP     |
| 152 | [248,46]   | [251,64]   | 20.071 | TIP     | TIP     |
| 153 | [223,48]   | [225,89]   | 43.485 | TIP     | TIP     |
| 154 | [465,53]   | [470,68]   | 17.071 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 155 | [339,58]   | [353,68]   | 21.657 | TIP     | TIP     |
| 156 | [442,58]   | [445,67]   | 10.243 | TIP     | TIP     |
| 157 | [383,92]   | [381,104]  | 13.657 | TIP     | TIP     |
| 158 | [330,98]   | [311,115]  | 27.799 | TIP     | TIP     |
| 159 | [240,104]  | [250,105]  | 11.243 | TIP     | TIP     |
| 160 | [453,108]  | [452,112]  | 4.414  | TIP     | TIP     |
| 161 | [214,119]  | [198,125]  | 18.485 | TIP     | TIP     |
| 162 | [441,132]  | [422,144]  | 27.728 | TIP     | TIP     |
| 163 | [254,140]  | [232,177]  | 47.87  | TIP     | TIP     |
| 164 | [501,167]  | [497,168]  | 4.414  | TIP     | TIP     |
| 165 | [372,200]  | [401,243]  | 66.527 | TIP     | TIP     |
| 166 | [150,216]  | [146,238]  | 23.657 | TIP     | TIP     |
| 167 | [134,217]  | [118,235]  | 26.385 | TIP     | TIP     |
| 168 | [327,219]  | [330,221]  | 3.828  | TIP     | TIP     |
| 169 | [91,232]   | [97,242]   | 13.071 | TIP     | TIP     |
| 170 | [427,233]  | [437,255]  | 26.142 | TIP     | TIP     |
| 171 | [399,255]  | [388,263]  | 18.314 | TIP     | TIP     |
| 172 | [352,285]  | [345,293]  | 11.485 | TIP     | TIP     |
| 173 | [86,286]   | [62,307]   | 34.456 | TIP     | TIP     |
| 174 | [183,290]  | [218,320]  | 49.184 | TIP     | TIP     |
| 175 | [41,302]   | [30,308]   | 13.485 | TIP     | TIP     |
| 176 | [452,371]  | [444,381]  | 13.899 | TIP     | TIP     |
| 177 | [21,375]   | [24,380]   | 6.243  | TIP     | TIP     |
| 178 | [69,392]   | [82,396]   | 14.657 | TIP     | TIP     |
| 179 | [140,395]  | [125,398]  | 17.071 | TIP     | TIP     |
| 180 | [2,397]    | [4,398]    | 2.414  | TIP     | TIP     |
| 181 | [229,416]  | [241,427]  | 20.657 | TIP     | TIP     |
| 182 | [6,454]    | [11,466]   | 19.243 | TIP     | TIP     |
| 183 | [250,470]  | [240,471]  | 11.243 | TIP     | TIP     |
| 184 | [34,474]   | [48,486]   | 21.556 | TIP     | TIP     |
| 185 | [195,484]  | [187,490]  | 11.071 | TIP     | TIP     |
| 186 | [117,490]  | [118,495]  | 5.414  | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 187 | [341,493]  | [349,510]  | 20.899 | TIP     | TIP     |
| 188 | [91,498]   | [108,499]  | 18.243 | TIP     | TIP     |
| 189 | [1,502]    | [13,511]   | 16.899 | TIP     | TIP     |